Viva & presentation prep

ReactorGenome

Speak these answers — not code.

App: https://che331.streamlit.app ů Report: docs/report/report.pdf ů Repo: https://github.com/Mayurdp2403/ReactorGenome

1 Thirty-second pitch

Thirty-second pitch — memorise this

CHE331 runs Batch, CSTR, PFR and RTD under fixed conditions. We write readings down and analyse later, so we never *see* what is going wrong inside the reactor during the lab. ReactorGenome is a web app on a laptop next to the rig. Students type the same readings they already take. The app fits kinetics, checks whether a run really reached steady state, lets you switch assumptions on and off, and then uses your fitted k to design plants the lab cannot build — series, recycle, jackets with multiple steady states. No new sensors. One app for all four labs.

2 The problem

What the lab does today

- One reactor, three flows, three temperatures, equal dilute NaOH + ethyl acetate.
- Readings on paper; analysis after the slot.
- "Why does X fall with flow?" answered in words, never by trying it.

What went wrong in our own CSTR data

- Conductivity still drifting when we called it steady.
- 28 °C gave higher conversion than 40 °C.
- We wrote "unstable readings" and moved on — no tool to diagnose it.

What we promised: a laptop web app (no new hardware) that takes lab-sheet readings, fits them, updates plots live, drops textbook assumptions one by one, and tries series / recycle / non-isothermal scale-up.

3 What you built (high level)

Layer	What it is	What to say
Front end	Streamlit (Python web UI)	*"Students open a browser; no desktop install."*
Maths	NumPy + SciPy	*"Same equations as Fogler / Levenspiel."*
Plots	Plotly	*"Interactive graphs; camera saves PNG."*
Design board	HTML/JS canvas	*"Drag feeds, tanks, tubes, splitters."*
Hosted at	Streamlit Cloud	https://che331.streamlit.app

One sentence: an engine (models only) + pages (Batch, CSTR, PFR, RTD, Design Studio).

Same layout on every experiment page

- **Left:** assumptions and run conditions
- **Middle:** animated reactor + live curves
- **Right:** results and diagnostics
- **Below:** readings table (like the lab sheet)
- **Tabs:** fits, assumption effects, sweeps, export

4 Chemistry & lab facts

Reaction: NaOH + ethyl acetate → sodium acetate + ethanol (saponification).

Why conductivity? OH^- is highly conductive; as NaOH is consumed, Λ falls:

$$X = \frac{\Lambda_0 - \Lambda}{\Lambda_0 - \Lambda_\infty}$$

Λ_0 = feed / initial; Λ_∞ = complete conversion.

App default: $\Lambda_\infty \approx 0.366\,\Lambda_0$ from limiting ionic conductivities (acetate replaces hydroxide). Manuals often type a fixed value (0.1 or 3.75 mS/cm).

If they ask for an insight

Λ_∞ that is too low **inflates** X and **deflates** k. Fitting from CSTR data gave $\approx 0.30\,\Lambda_0$ — close to the ion estimate — and k at 31 °C rose from ~ 3.2 to ~ 8.5 L/molůmin. That is the best "what did the app find that the report missed?" answer.

Usual rate law: $-r_A = k\,C_A C_B$ (equimolar $\Rightarrow k\,C_A^2$).

Damköhler: $\mathrm{Da} = k\,C_{A0}^{n-1}\,\tau$ — how fast reaction is vs how fast fluid moves through.

5 Core concepts (speakable)

5.1 CSTR

Ideal: outlet = tank contents. Design (2nd order, equimolar): plot $y = X/(1-X)^2$ against τ; slope $\rightarrow k$. Settling needs several τ_r, not "one τ". Bypass / dead volume lower X; tanks in series raise X toward plug flow.

5.2 PFR

Ideal: plug flow along the coil. Design: $y = X/(1-X)$ vs τ. Sample delay in the beaker looks like a huge free intercept; the app models it as an explicit delay. Alternatives: dispersion (Pe), laminar, recycle, adiabatic.

5.3 Batch

$\kappa(t) \rightarrow C_A(t) \rightarrow$ fit k. Differential method: slope of $\ln(-\mathrm{d}C/\mathrm{d}t)$ vs $\ln C$ is order n. Our data: $n \approx 1$, not 2 — same as the report's differential method. Say: mixing dominates early points; after the window, first order wins by AIC.

5.4 RTD

Tracer $\rightarrow E(t)$, t_m, σ^2. $N = t_m^2/\sigma^2$. Ideal CSTR $\Rightarrow N = 1$. Packed bed: high Pe. If you cut the tail early, σ^2 is too small and N looks too high ($\sim 1.5 \rightarrow \sim 1.2$ with a tail).

5.5 Arrhenius

$\ln k = \ln A - E_a/(RT)$. App gives 95% intervals on E_a and A.

5.6 Multiple steady states

Heat generation vs jacket removal can cross three times (two stable, one unstable) when the exotherm is strong and cooling is weak. Dilute lab feeds barely heat; the app's what-if uses concentrated feed + weak jacket.

6 How you know it works

1. Closed-form limits (ideal CSTR/PFR, tanks→plug flow, RTD moments, recycle) — errors 10^{-10} to 10^{-16}.
2. 2026 lab reports with *their* conventions: k and E_a match to four figures.
3. Report readings are **not** built into the app (empty tables on start).

CSTR E_a	25.07 kJ/mol	(k: 3.16 / 5.85 / 7.44 at 31/46/60 řC)
PFR E_a	12.70 kJ/mol	(k: 11.74 / 13.75 / 15.73 at 40/50/60 řC)
Design Studio demo	CSTR then PFR $\Rightarrow X \approx 0.545$, balance $< 10^{-8}$	

7 Every figure — what it is and why

Use with `report.pdf`. If they point at a screenshot, say this.

7.1 Home

Landing cards for each experiment + Design Studio. Shows one app for all four labs.

7.2 CSTR overview

Assumption switches, animated tank, KPIs, fitted E_a/k. Proves the live digital-twin layout.

7.3 CSTR readings

Lab-sheet table (T, flow, τ, $\Lambda_{1\ldots4}$) + Λ_0 per temperature. Matches the manual's observation table — proposal point "enter the readings".

7.4 CSTR results

E_a, A, $k(T)$, design plots, Arrhenius, X vs flow/T, single-run k vs τ, parity, residuals. Everything a lab report needs. **Single-run k vs τ:** flat = consistent; sloping = bad Λ_∞, unsteady runs, or delay.

7.5 CSTR predictions

Sliders for T, C_{A0}, τ. Orders 1–3 all fit at the lab C_{A0}; they separate only if concentration changes — that is how order is told apart.

7.6 CSTR assumptions

Bar chart: ideal vs bypass / dead volume / 3 tanks / stagnant / reversible. Proposal: flip assumptions and *see* the cost.

7.7 CSTR energy

Heat generation vs removal (three crossings) + phase portrait. Scale-up / runaway the dilute lab cannot show.

7.8 PFR overview / results / flow models

Coil animation and kinetics ($E_a = 12.70$); full results grid; same k under plug / dispersion / laminar with points overlaid. Non-ideal tube without rebuilding the rig.

7.9 Batch overview / fit

$C_A(t)$ and fitted k; integral + differential plots, AIC ranking. Order is a choice, not an assumption.

7.10 RTD overview / all runs / model fits

$E(t)$ + tail warning; four-run overlays; tanks-in-series vs dispersion ranked by error. Mixing quality $\rightarrow N$, Pe, conversion.

7.11 Design Studio

Drag-and-drop plant (e.g. CSTR then PFR), stream/unit tables, $X = 0.545$. "What the lab cannot do."

7.12 Export

ZIP (HTML + CSV + plots + settings). Also: Project $\rightarrow$ Save writes a `.reactorgenome.json` another student can Load — share a Design Studio plant.

8 How students and TAs use it in the 3-hour lab

3-hour lab — one sentence

Laptop beside the rig $\rightarrow$ type readings live $\rightarrow$ check settled $\rightarrow$ fit k $\rightarrow$ Design Studio $\rightarrow$ export ZIP + save JSON.

8.1 Before the slot (TA)

Open the live URL (or `run.bat` locally). Set project name = group. Remind: type what the manual already asks. Optional: load a previous group's JSON.

8.2 Timing

Time	Phase	Students	TA focus
0–15 min	Setup	V, C_{A0}, Λ_∞ mode, T	Concentrations match bottles
15–40	Run 1	Start flow; type when due	Don't average a climbing Λ
during	Live check	Watch transient / status	"Has it settled?"
40–100	More runs	Change Q or T	Point at k vs τ if values drift
100–130	Assumptions	Toggle bypass / dispersion	"Which assumption moves X most?"
130–160	Beyond rig	Design Studio with fitted k	Series / recycle / same volume
160–180	Wrap-up	Export ZIP + Save JSON	Collect JSON if continuing at home

8.3 TA coaching lines

- **CSTR:** low-T $X >$ high-T $X \Rightarrow$ readings still rising. Falling k with $\tau \Rightarrow \Lambda_\infty$ too low.

- **PFR:** nonsense free-line $k \Rightarrow$ through-origin fit or turn on sample delay.
- **Batch:** set window past mixing; if $\kappa < \kappa_\infty$, fix typed κ_∞.
- **RTD:** record to baseline or use tail correction; tank $N \sim 1$–2 vs bed high Pe.

8.4 Design Studio (last half hour)

After a fit, second-order k and E_a are shared. Load preset "CSTR then PFR" ($X \approx 0.545$). Try recycle / bypass. Optional jacketed strong-exotherm what-if for multiple steady states. Save JSON to share the plant.

Board conceptually: feeds, CSTRs, PFRs, beds, splitters, products; mass balances; recycle iterated to convergence; mass-balance error reported. Tabs: Streams, Profiles, Levenspiel, Sweep, Compare rigs, Export.

9 Likely viva Q&A

Q: What is ReactorGenome?

A browser digital twin for the four CHE331 reactor labs. Type existing readings; get fits, diagnostics, assumption checks, and a design sandbox.

Q: Why not Aspen / COMSOL?

Heavy, licensed, not built around this lab sheet and conductivity workflow. We needed something a group opens in minute one of a three-hour slot on a laptop.

Q: What is new vs Excel?

Live transient and steady-state judgement, ion-based Λ_∞, non-ideal models, shared k across rigs, flowsheet design, full export — while typing, not after the lab.

Q: How is conversion obtained?

From conductivity with Λ_0 and Λ_∞. Default from ion conductivities; typed value or fit from data also available.

Q: How do you get k and E_a?

Design plot (y vs τ) at each T, then Arrhenius across temperatures, with 95% intervals.

Q: What is Damköhler number here?

$\mathrm{Da} = k\, C_{A0}^{n-1}\, \tau$. Large Da $\Rightarrow$ high conversion for a given reactor type.

Q: CSTR E_a 25 vs PFR 12.7 — why different?

Different groups, different Λ_∞ conventions, possible sample delay and scatter. The app matches each report's own method; honest comparison needs the same rules — that is what Compare rigs is for.

Q: How do you know the maths is right?

Analytic limits + reproducing 2026 lab-report k and E_a to four significant figures; 130+ automated tests.

Q: What did the app reveal that reports missed?

(1) $\Lambda_\infty \approx 0.3\,\Lambda_0$ fits CSTR better than 0.1 mS/cm. (2) Temperature inversion was largely unsteady averaging. (3) RTD $N \approx 1.5$ was mostly a cut-off tail. (4) Batch differential order ≈ 1.

Q: Bypass vs dead volume?

Bypass: fraction of feed never enters $\Rightarrow$ dilutes product, lowers X. Dead volume: part of V stagnant $\Rightarrow$ smaller effective τ, lowers X. Both on Assumption effects.

Q: Multiple steady states — when?

Strong exotherm + limited cooling (finite UA). Generation and removal can intersect three times; middle state unstable.

Q: Can groups share work?

Project Save $\rightarrow$ `.reactorgenome.json` $\rightarrow$ Load project. Design Studio board and all tables restore.

Q: Limitations (be honest)?

Dead volume / stagnant zones are compared more than fully fitted (bypass can be fitted). Energy balance is a what-if for dilute labs. Only second-order k is shared across pages. Laptop-first; narrow phone tabs clip.

Q: How would a TA use it tomorrow?

Open today's experiment page, watch status while groups type, catch unsteady CSTR early, then push them into Design Studio with their own k in the last hour.

10 Five-minute demo script

1. **Problem** (45 s) — drifting CSTR readings; answers only in words.
2. **Demo Home** $\rightarrow$ **CSTR** (90 s) — readings table $\rightarrow$ Results (E_a, plots).
3. **Assumption effects** (30 s) — flip bypass, show bar chart.
4. **Design Studio** (60 s) — CSTR then PFR, $X = 0.545$.
5. **Lab use + JSON** (45 s) — 3-hour workflow; Save/Load.
6. **Close** (30 s) — validated against reports; live at che331.streamlit.app.

Keep `report.pdf` open on figures if Wi-Fi fails.

Last look before you walk in

- ✓ **What:** digital twin for Batch / CSTR / PFR / RTD + Design Studio.
- ✓ **Input:** same readings as the manual.
- ✓ **Output:** X, $k(T)$, E_a, diagnostics, assumption costs, plants, ZIP.
- ✓ **Key insight:** bad Λ_∞ and unsteady averages beat "bad kinetics".
- ✓ **Lab use:** type live $\rightarrow$ check settled $\rightarrow$ fit $\rightarrow$ design $\rightarrow$ export.

✓ **Stack if asked:** Python, Streamlit, NumPy/SciPy, Plotly; Streamlit Cloud.

✓ **Not claiming:** new hardware, phone-perfect UI, or Aspen for industry plants.

Speak the pitch. Show Results + Design Studio. Defend the rest from this sheet.